

Subject Name: Plant Tissue and Cell Culture Laboratory

Subject No: BT29202

Prerequisite: Plant Tissue and Cell Culture (BT20202)

Objective

The content targets full hands-on practices on all aspects of plant tissue and cell culture. Individual experiments are given to all students to build confidence. The students learn about laboratory design, details of instruments in creation of a new laboratory, follow up all aspects of media preparation, culture initiation, organogenesis and mass clonal propagation. They are adequately exposed to culture apparatus and culture systems that lead to cost effective plant propagation *in vitro*.

L-T-P 0-0-3

Credit 2

Content

- 1 Preparation of media (6 hours)
- 2 Demonstration of instruments and their uses (6 hours)
- 3 Culture initiation (Callus culture) (3 hours)
- 4 Micropropagation of herbaceous & woody plants (6 hours)
- 5 Cell suspension culture initiation (6 hours)
- 6 Comparative study of solid & liquid media for micropropagation using different culture apparatus(6 hours)
- 7 Cell suspension culture in airlift bioreactors (6 hours)
- 8 Protoplast isolation & fusion (3 hours)
- 9 Somatic embryogenesis of sandalwood (3 hours)
- 10 Artificial seed production (3 hours)

Text books-

1. Plant Cell, Tissue and Organ Culture- Fundamental Methods by Gamborg, Oluf, Phillips, Gregory (Eds.), Publisher- Springer.ISBN 978-3-642-79048-5
2. Plant Tissue Culture: Theory & Practice by Bhojwani & Razdan Publisher-Elsevier eBook ISBN: 9780080539096
3. Plant Biotechnology: Laboratory Manual by H. S. Chawla Publisher- Oxford & IBH ISBN-13 : 978-8120416130

4. Methods in Plant Molecular Biology & Biotechnology by BR Glick & JE Thompson, Publisher- CRC Press, ISBN 0-8493-5164-2

Further reading references-

1. Kannan et al. Induction of adventitious roots from leaf explants of *Morinda coreia* Buch. and ham. : an important dye yielding plant. Plant Cell, Tissue and Organ Culture (PCTOC) volume 145, pages 457–460 (2021).
2. Tian et al. Riboflavin fortification of rice endosperm by metabolic engineering. Plant Biotechnology Journal. <https://doi.org/10.1111/pbi.13615>.
3. Sharma & Negi. Production and Challenges of Plant based Vaccines. Annals of R.S.C.B., 25, 3625-3639 (2021).